Card Create

User Guide

Guide for Card Create version 3.0

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Guide for Card Create version 3.0
(CardCreate3)

Phil Lockett

Handmade Books

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1 Introduction

A while back I was looking for software that would create images representing playing cards. I found a bash script that would generate cards with images on them using the 'convert' command line utility (now 'magick') from ImageMagick. It was not what I wanted but it did motivate me to write some C++ code to generate a bash script which itself could generate each card the way I wanted. This software gave more control and flexibility for creating playing card images. I called this software 'cardgen'.

I then re-wrote it in java to be independent of the 'convert' command and use a GUI instead of command line parameters. To differentiate this java version, I called it 'CardCreate'. Then, to learn JavaFX I rewrote it again, calling it 'CardCreate2'. I then added embedded SVG images and called it 'CardCreate3'.

To compose the playing cards, 'CardCreate3' (like it's predecessors) can use separate .png image files for the faces of the court cards, the indices and the pips. These images are kept in separate directories, with a sub-directory for each style. This allows for different decks of cards to be generated from different combinations of styles of these component parts. We will see examples of this later.

The embedded SVG images added to 'CardCreate3' provide standard images for indices, pips and court faces. These SVG images provide an alternative to the .png files and enables easy colour changes for custom decks.

The intention for 'CardCreate3' is to quickly prototype playing card decks without having to create and position every component of every card individually.

2 Installing the software

2.1 System requirements

The system requirements for using 'CardCreate3' are:

- Java Run-time Environment
- Maven
- git
- tar utility

To get started we first need a java runtime environment installed. To check if java is installed launch a terminal/console and run '**`java -version`**'. You should see something similar to the following:

```
$ java -version
java version "15.0.1" 2020-10-20
Java(TM) SE Runtime Environment (build 15.0.1+9-18)
Java HotSpot(TM) 64-Bit Server VM (build 15.0.1+9-18, mixed mode, sharing)
```

If not, go to the Oracle web site (https://www.java.com/en/download/) and install the version appropriate to your operating system.

Similarly check if 'maven' is installed:

```
$ mvn --version
Apache Maven 3.6.3 (cecedd343002696d0abb50b32b541b8a6ba2883f)
Maven home: K:\apache-maven-3.6.3
Java version: 15.0.1, vendor: Oracle Corporation, runtime: C:\Program Files\
Java\jdk-15.0.1
Default locale: en_CA, platform encoding: Cp1252
OS name: "windows 10", version: "10.0", arch: "amd64", family: "windows"
```

If maven is not installed go to the Apache web site (https://maven.apache.org/install.html) and follow the instructions.

Finally, check if 'git' is installed:

```
$ git --version
git version 2.35.1
```

If not go to the git web site (https://git-scm.com/book/en/v2/Getting-Started-Installing-Git). Note other git options are available.

In the following examples the tar utility is used to extract the tar file, but most file explorers will extract the files also and can be used if preferred.

2.2 Clone and Build the 'CardCreate3' application

For the following example we will use '/cygdrive/k/Users/Phil/Work/Playing Cards' as a parent directory. We will clone 'CardCreate3' as follows:

```
$ cd '/cygdrive/k/Users/Phil/Work/Playing Cards'
```

```
$ git clone https://github.com/PhilLockett/CardCreate3.git
```

2.3 Set up the 'CardWork' environment

The 'CardCreate3' package comes with java source code and a 'CardWork' environment in which to use it. This environment provides some component images to get you started. Note that this set up is only required once, unless you want multiple copies of the environment, in which case repeat the steps for each copy.

For the following commands we will again use '/cygdrive/k/Users/Phil/Work/Playing Cards' as a parent directory, there we will copy 'CardWork.tar.gz' into it, then execute the following commands:

```
$ cd '/cygdrive/k/Users/Phil/Work/Playing Cards'
$ cp ./CardCreate3/CardWork.tar.gz .
$ tar zxf CardWork.tar.gz
$ rm CardWork.tar.gz
$ cd CardWork/
$ ./setup.sh
```

This creates the following file structure containing numerous .png image files:

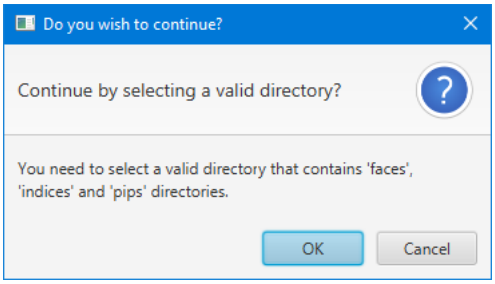
```
└──CardWork
    ├──boneyard
    ├──faces
    │   ├──1
    │   ├──2
    │   └──Rouen
    ├──indices
    │   ├──1
    │   ├──2
    │   ├──3
    │   └──Rouen
    └──pips
        ├──1
        ├──2
        ├──3
        └──Rouen
```

3 Running ‘CardCreate3’

To run ‘CardCreate3’ change to the ‘CardCreate3’ directory and execute the maven command:

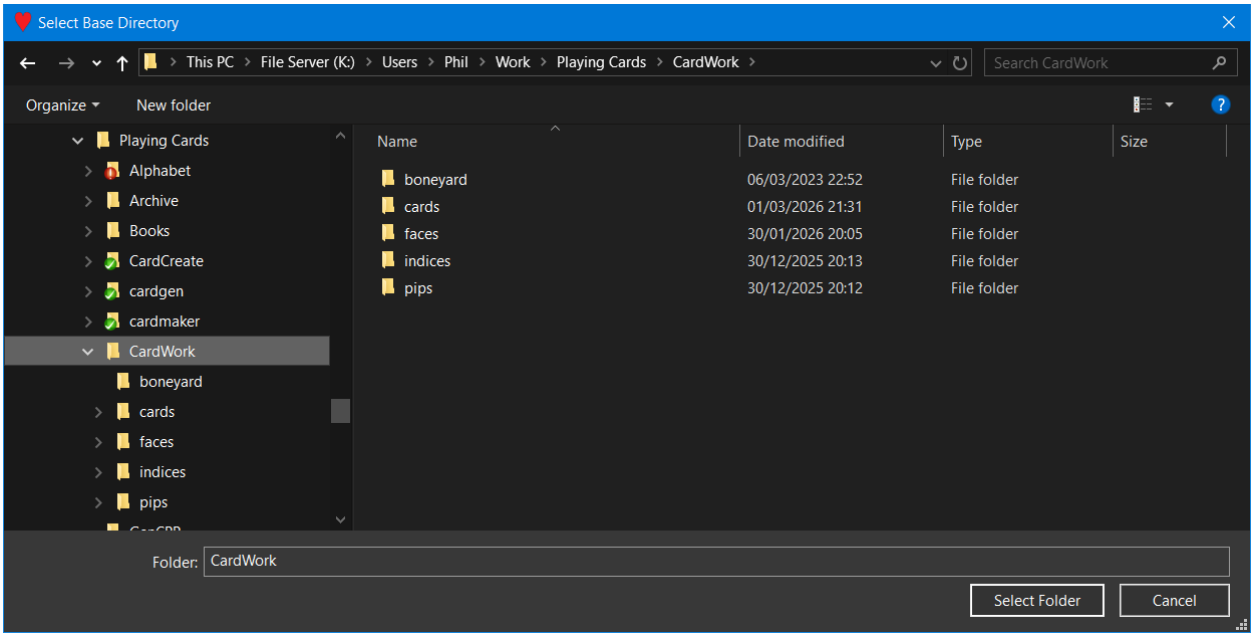
```
$ cd '/cygdrive/k/Users/Phil/Work/Playing Cards/CardCreate3'
$ mvn clean javafx:run
```

When launched for the first time, the following ‘continue’ dialogue will be shown:



Selecting ’Cancel’ will shut down the application.

Selecting ‘OK’ will launch the following directory selection dialogue which allows you to navigate to a directory that contains the sub-directories: ‘faces’, ‘indices’ and ‘pips’ such as the one created above in the ‘CardWork’ directory:



Base Directory selection window

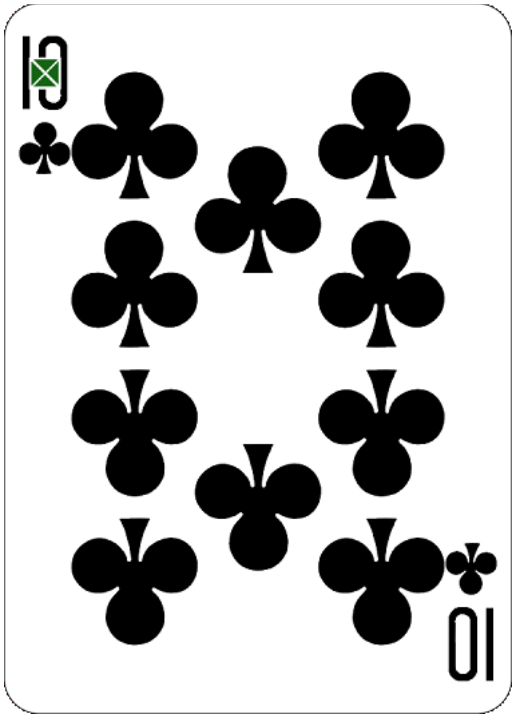
If ‘Cancel’ is selected at this point, or a directory which does not contain the necessary sub-directories is opened, you are returned to the ‘continue’ dialogue, where you can ‘Cancel’ to exit the application.

If a valid directory is selected, the application and the card ‘Sample’ windows are launched:

Main Application Control Window - Primary Tab

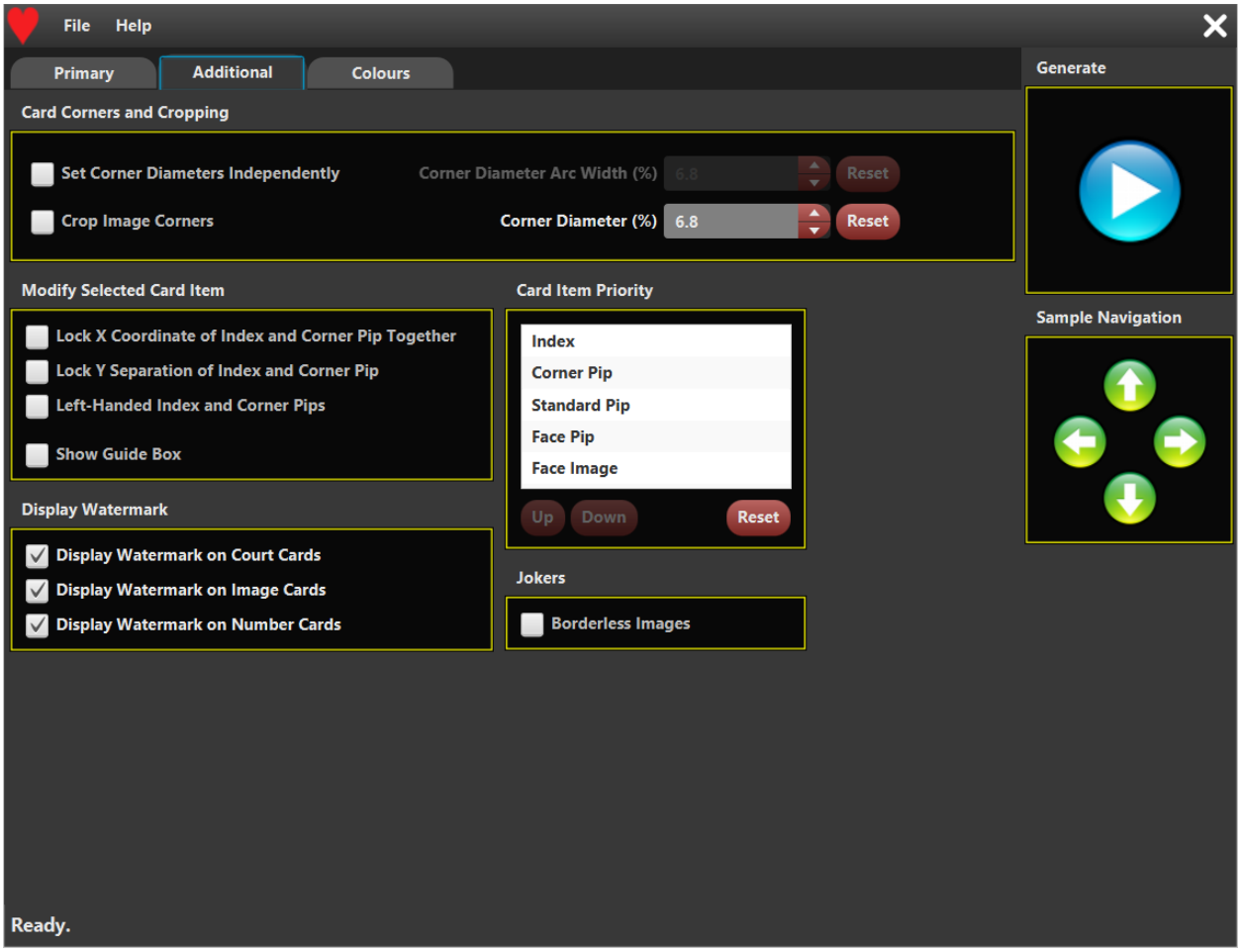
Note: now a ‘Base‘ directory has been selected, it is not necessary to select one again and subsequent ‘CardCreate3’ application launches will automatically use the directory already selected. However, other valid directories can be selected using the ‘Browse…’ button, and these directories can then be switched between using the pull-down.

Card composition takes place in the ‘CardWork’ directory. Note that this ‘CardWork’ directory can be named as needed, but the sub-directories it contains should not be renamed as ‘CardCreate3’ has these hard-coded and looks for them explicitly.



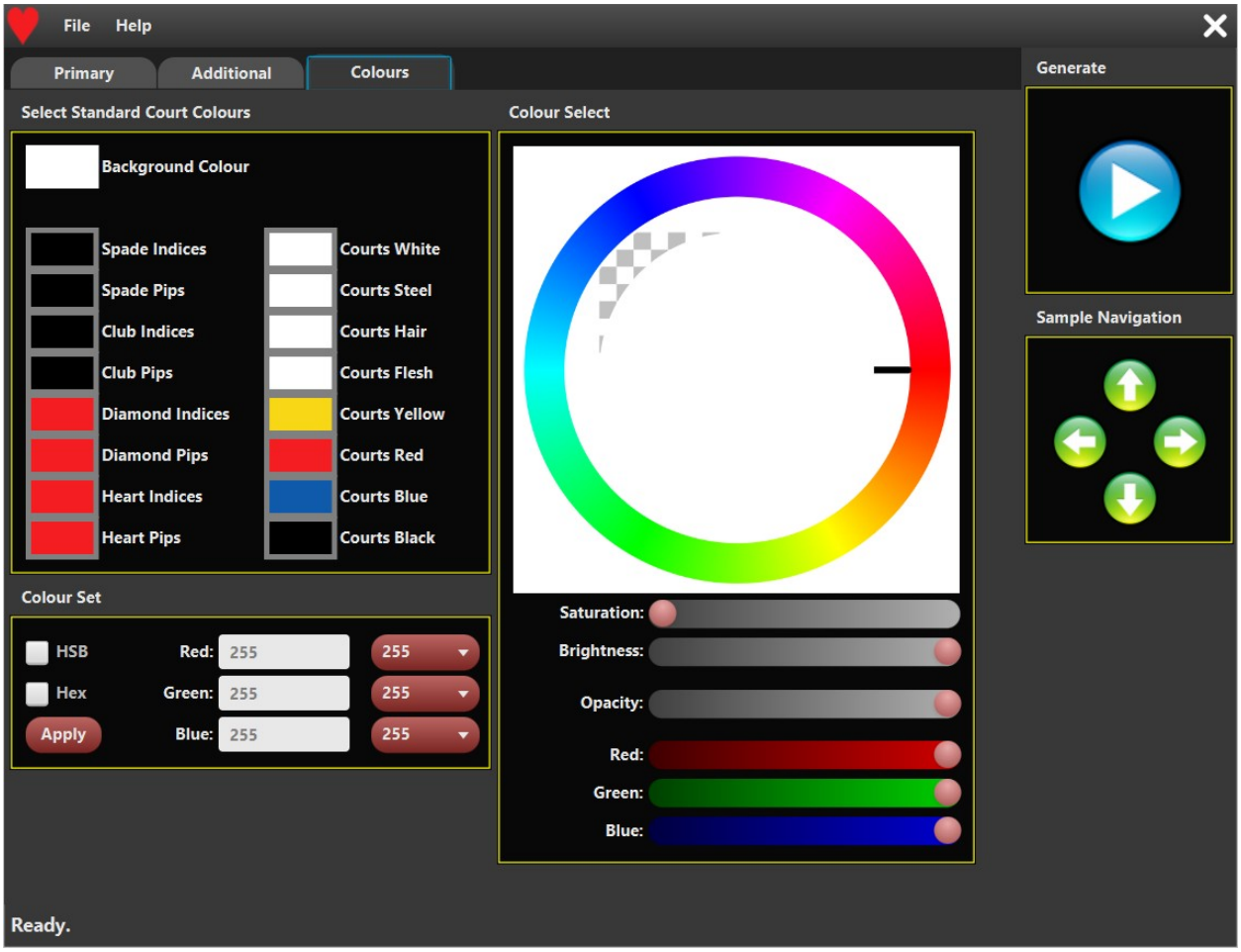
Sample Window

There are a few more options that can be set by clicking on the ‘Additional’ tab of the main control panel:



Main Application Control Window - Additional Tab

Finally there are colour options for the Standard SVG images under the 'Colours' tab of the main control panel:



Main Application Control Window - Colours Tab

All of these options are explained below.

3.1 'CardWork' Environment breakdown

The 'CardWork' directory provides the following sub-directories:

- boneyard
- faces
- indices
- pips

3.1.1 boneyard

The 'boneyard' contains additional image files used for generating default Jokers. The 'CJoker.png' 'DJoker.png', 'HJoker.png' and 'SJoker.png' images files should be copied into the selected 'indices' directory and used as required.

3.1.2 faces

The ‘faces’ directory contains sub-directories of images used for the different styles of face cards. Typically, these are the court cards plus the ace of spades image. The only difference between style 1 and 2 is the colour of the ace of spades. The Rouen sub-directory is a re-imagining of an old-style deck of cards but with added ace of spades and jokers.

3.1.3 indices

The ‘indices’ directory contains sub-directories of images used for the different styles of indices. The only difference between style 1 and 2 is the colour of the club indices, black has been changed to white. Sub-directory 3 contains four sets of indices. Typically, a deck only uses red and black indices, but this example uses different colours for each suit.

Note: if images for the spades indices are not found, clubs will be used for the spade cards. Similarly, if indices for hearts are not found, indices for diamonds will be used.

3.1.4 pips

The ‘pips’ directory contains sub-directories of images used for the different styles of pips. The only difference between style 1 and 2 is the colour of the club and spade pips, black has been changed to white. Sub-directory 3 contains only four pip images. These are used for both the standard pips and the corner pips. Optionally you could add a set of small pips to be used as the corner pips. More on this below.

All of these images are used to compose cards by resizing and positioning them.

Note: if the selected ‘faces’ sub-directory contains an image for a number card, the image is used if ‘File Override’ is selected, otherwise the standard pips arrangement is used.

3.1.5 cards

In addition a new ‘cards’ directory is added to these directories when card images are first generated and a sub-directory created, named using the ‘Output Directory’ option. By default the output sub-directory is named after the faces sub-directory, but this can be changed. Card images are created inside this sub-directory.

4 Anatomy of a Playing Card

'CardCreate3' composes playing cards from five card items and allows the position and size to be modified, but in a manor consistent with all individual instances of the item. For example, changing the position and size of the index changes both the top-left and bottom-right index. The five card items are:

- Index
- Corner Pip
- Standard Pip
- Face Pip
- Face Image

The order listed above indicates the order in which they are displayed on the card. So, if they overlap, the Index is displayed on top of the Corner Pip which is on top of the Standard Pip etc. If necessary, this order can be changed using the 'Card Item Priority' panel in the 'Additional' tab.

All image files must be .png files and it is highly recommended to use transparency.

4.1 Index

The index is the letter or number that appears in the two corners of the card. Images representing the indices are found in the 'indices' directory. See the 'Notes' section (at the end of this document) for more details on the file naming convention and avoiding unnecessary file duplication. Example:

There is only one control for the Jokers which is to control whether a border is used around the image or not, which is found on the 'Additional' tab. Generating jokers also relies on images being placed in the 'faces' directory as needed.

The Joker card can be augmented with the joker text, as an image, placed in the 'indices' directory. A sample of the Joker text images are available in the 'boneyard' directory and should be copied or created and named as needed. This text image is added to the corners of the generated joker cards.

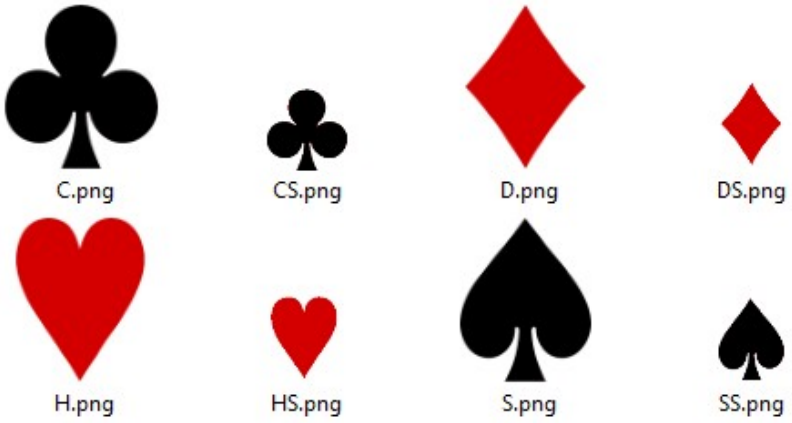
The file naming convention is also based on the initial letter of the suit followed by 'Joker'.

4.2 Suit Symbols (pips)

The pip is the suit symbol displayed underneath the index and repeated for the number cards as well as on the court cards. Images representing the pips are found in the 'pips' directory. See the 'Notes' section (at the end of this document) for more details on the file naming convention.

4.2.1 Corner Pip

The Corner pip is the suit symbol displayed underneath the index. The naming convention uses the initial letter of the suit. These are used for the standard pip, the corner pip and the face pip on the court cards. Optionally an additional set of images can be provided allowing for the option to have simpler suit signs for the smaller corner pip. If the smaller image is available it is used, if not the standard pip image is used. The example below shows images for both the standard pip and the corner pip.



4.2.2 Standard Pip

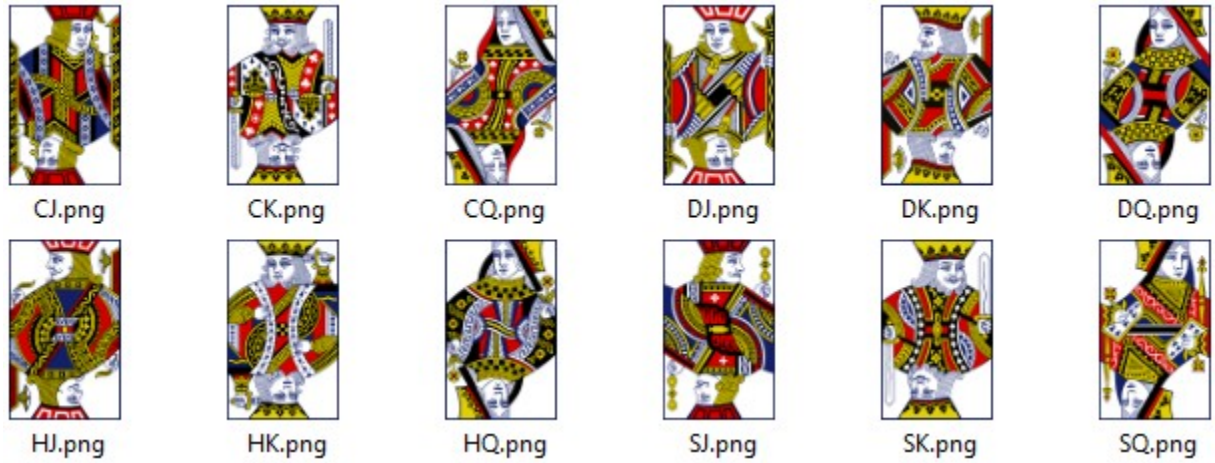
The Standard pip is the suit symbol displayed in the centre of the number cards (Aces through 10s). The images representing the Standard pips are found in the 'indices' directory.

4.2.3 Face Pip

The Face pip is the suit symbol displayed in the corner of the face image of the court cards (Jacks, Queens and Kings) and can use the same image files as the Standard pips. Optionally an additional set of images can be provided specifically for the face pip. This allows for the option to have alternate suit signs for the court cards. If the face pip image is available it is used, otherwise the standard pip image is used. The Face pip is typically overlaid on top of the face image of the court cards, but only if 'Display Face Pip' is checked in the 'Display Card Items' panel (demonstrated below).

4.3 Face Image

The Face image is the image that appears in the centre of the card. Images representing the faces are found in the 'faces' directory. See the 'Notes' section (at the end of this document) for more details on the file naming convention.



Typically, these images represent the court cards, but may also include an image for any other card, such as the ace of spades (SA.png). Notice that these court card face images do not include the suit symbol which is controlled by the Face pip item and is overlaid on the face image.

Face images are not restricted to the court cards and ace of spades. Any card can be represented by a face image using the naming convention of the initial letter of the suit followed by the index as long as the 'File Override' check-box is selected.

Joker images can also be provided, using the naming convention of the initial letter of the suit followed by 'Joker'. Simply create images for 'CJoker.png', 'DJoker.png', 'HJoker.png' and 'SJoker.png' as needed.

5 ‘CardCreate3’ Controls

‘CardCreate3’ uses two windows, the main application control window labelled ‘Playing Card Generator 3.0’ and the card ‘Sample’ representation window, which also provides both Mouse and Keyboard controls. The main application control window provides the following control options:

- Pull-Down Menus
  - File
    - Open Base Directory...
    - Load Settings
    - Save Settings
    - Close
  - Help
    - About
    - Help
- Primary Tab
  - Input Directories
  - Output Directory
  - Colour Theme
  - File Override
  - Card Size
  - Card Item Selection and Modification
    - Select Card Items to Display
    - Select Card Item to Modify
    - Modify Selected Card Item (%)
- Additional Tab
  - Card Corners and Cropping
  - Modify Selected Card Item
  - Display Watermark
  - Card Item Priority
  - Jokers
- Colours Tab
  - Select Standard Court Colours
  - Colour Select
  - Colour Set
- Sample Navigation
- Generate
- Status Line

5.1 Pull-Down Menus

The main control window includes pull-down menus which provide alternate ways of triggering certain actions as well as providing useful information under the 'Help' option.

5.1.1 File → Open Base Directory...

Valid 'Base' directories can be selected by using the 'Open Base Directory…' option which launches a directory selection dialogue. A base directory must contain 'faces', 'indices' and 'pips' sub-directories which themselves must each contain at least one sub-directory to be considered valid.

5.1.2 File → Load Settings

If a settings file has previously been created the 'Load' option is available and can be used to load those settings.

5.1.3 File → Save Settings

The 'Save' option saves the current settings to the output directory.

5.1.4 File → Close

This option closes the application.

5.1.5 Help → About

Selecting this option launches a pop-up that indicates the release number of the Playing Card Generator and a synopsis of its function.

5.1.6 Help → Help

Selecting this option launches a pop-up that directs the User to this guide.

5.2 Main Application Control Window - Primary Tab

5.2.1 Input Directories

The 'Input Directories' control pane contains the 'Base' directory combo-box which was initially populated during the set-up performed above. The 'Base' directory must contain the three sub-directories; 'faces', 'indices' and 'pips' which themselves must each contain at least one sub-directory to be considered valid.

The 'Browse' button duplicates the 'Open Base Directory…' option of the pull-down menu described above.

The 'Base' directory combo-box maintains all the previously selected 'Base' directories and can be changed between them using the pull-down button. These 'Base' directories are persisted between sessions.

The 'Use Standard Faces', 'Use Standard Indices' and 'Use Standard Pips' check-boxes enables the use of embedded SVG images which have the advantage of being colour modifiable. Deselecting these allows the corresponding 'Face', 'Index' and 'Pip' choice-boxes to be utilized.

The 'Face', 'Index' and 'Pip' choice-boxes are populated using the sub-directory names found in their corresponding directory. The 'CardWork.tar.gz' package provides some example images to use. The

‘faces’ directory contains ‘1’, ‘2’ and ‘Rouen’ sub-directories. The ‘indices’ and ‘pips’ directory both contain ‘1’, ‘2’, ‘3’ and ‘Rouen’ sub-directories. Each of these sub-directories contain relevant .png image files that can be used to style the cards. The cards are composed by independently selecting styles for the ‘faces’, ‘indices’ and ‘pips’ images.

The ‘Load’ and ‘Save’ buttons duplicate the ‘Load’ and ‘Save’ options of the pull-down menu described above.

5.2.2 Output Directory

By default, the cards are generated in a sub-directory under the ‘cards’ directory with the same name as the faces directory. So, if the selected ‘faces’ directory is ‘1’, the ‘Output Directory’ is ‘1’ and the cards are generated in the ‘…/cards/1’ sub-directory.

However, the ‘Output Directory’ can be set manually by clicking the ‘Manual’ button and entering the name of the directory to use in the text box. **Do not use spaces in the name.** For example, if the name ‘test’ is entered the output will be found in ‘…/cards/test’. The ‘Manual’ button is a toggle button, so clicking it again will revert to using the name of the selected ‘faces’ directory once more.

5.2.3 Colour Theme

The ‘Colour Theme’ choice-box is used when using “Standard” faces, Indices and/or Pips. It enables multiple colours to be set by selecting one of the colour themes. Any of these colours can then be modified using the ‘Colours’ tab.

5.2.4 File Override

The ‘File Override’ check-box is used when standard pips are desired, but a specific number card may need to use a .png image file. For example, the ace of spades may have a custom design requirement which is provided by the file ‘SA.png’. This file should be placed in the ‘faces’ directory currently selected by the ‘Face’ choice-box, even though “Standard” faces may be being used and the ‘Use Image file if available’ check-box should be selected. This is specific to non face cards only.

5.2.5 Card Size

Playing cards are typically manufactured in two standard sizes, Poker size being 2½ by 3½ inches and Bridge size being 2¼ by 3½ inches. By default, the card size is selected as Poker with pixel values of 380 and 532. The ‘Card height’ can be adjusted using the spinner and the ‘Card width’ is automatically adjusted to maintain the aspect ratio. If Bridge size cards are selected, the ‘Card width’ is automatically adjusted to maintain the aspect ratio for Bridge cards. If ‘Free’ is selected, the card width and height can be adjusted independently.

If MPC is selected the output is adjusted for an MPC compatible Poker size card image which satisfies 2½ by 3½ inches at 300DPI with a 36 pixel print bleed border. These images can be directly dropped into the Poker Card design stage at https://www.makeplayingcards.com/.

The ‘Reset’ buttons return the pixel count to the default values.

5.2.6 Card Item Selection and Modification

5.2.6.1 Select Card Items to Display

The ‘Anatomy of a Playing Card’ section above lists the five card items as ‘Indices’, ‘Corner Pip’, ‘Standard Pip’, ‘Face Image’ and ‘Face Pip’. The ‘Display Card Items’ controls which card items are displayed.

Note: the ‘Face Pip’ images are only displayed on the court cards and that ‘Standard Pips’ are displayed if a corresponding face image file is not being used.

5.2.6.2 Select Card Item to Modify

The ‘Select Card Item’ controls which card item will be modified using either the ‘Modify Selected Card Item (%)’ control or the ‘Sample’ window.

The ‘Change Face Pip’ radio button can only be selected if ‘Sample’ is showing a court card. The ‘Change Face Border’ or ‘Change Standard Pip’ radio button is selectable based on whether ‘Sample’ is showing a card with a face image or standard number card pips.

Additionally, if the check box in the ‘Select Card Items to Display’ pane is unchecked the corresponding radio button will not be available at all.

5.2.6.3 Modify Selected Card Item (%)

The ‘Modify Selected Card Item (%)’ control pane allows the selected card item to be moved or resized. This can also be done using the ‘Sample Controls’ in the ‘Sample’ window, but these controls provide a more precise way of doing this.

The spinner values are percentages (%) and take 0.05% steps.

The ‘Size (%)’ value is the size of the selected card item as a % of card height.

The ‘X Centre (%)’ value is the X value of centre of the selected card item as a % of card width, 0 at the left, 100 at the right.

The ‘Y Centre (%)’ value is the Y value of centre of the selected card item as a % of card height, 0 at the top, 100 at the bottom.

When the ‘Select Card Item’ is changed the values on the spinners are updated to reflect this, for example the default value for the height of the corner pip is 7.5% whereas the default value for the height of the standard pip is 18.5%.

The ‘Reset’ buttons return the current value on the spinner to the default value for the ‘Select Card Item’.

When the ‘Select Card Item’ is the ‘Face Border’, you are not changing the position, but the X and Y borders instead and the height is no longer applicable. The face image is always centrally positioned.

The ‘Keep Image Aspect Ratio’ check box only applies to face images. When this is selected the image is proportionally drawn. When the check box is deselected, the image is stretched to fit within the X and Y borders.

5.3 Main Application Control Window - Additional Tab

Clicking on the 'Additional' tab shows additional settings.

5.3.1 Card Corners and Cropping

The 'Corner Diameter (%)' value adjusts the rounded corners as a % of card height (default: 6.8, i.e. 6.8% of the configured card height). Typically, this is sufficient, however, the Arc Width diameter and Arc Height diameter can be set independently if the 'Set Corner Diameters Independently' checkbox is selected.

The 'Crop Image Corners' checkbox removes any part of an image that is outside of the card. This is useful for full card images, however it takes some time so it is optionally applied and not always necessary.

5.3.2 Modify Selected Card Item

The 'Lock X Coordinate of Index and Corner Pip Together' checkbox makes the index image and the corner pip image stay vertically aligned with each other.

The 'Lock Y Separation of Index and Corner Pip' checkbox ensures the index image and the corner pip image maintain the same vertical distance apart.

When the 'Left-Handed Index and Corner Pips' checkbox is selected indices and corner pips are displayed in all four corners of the card which is beneficial to left handed players.

The 'Show Guide Box' helps indicate the card items being modified and is especially useful for modifying face images when the aspect ratio is maintained. When this is selected a box is drawn centrally on the 'Sample' window.

5.3.3 Display Watermark

If a 'Watermark.png' file is found in the 'Face' style directory, it is displayed behind all other images on all the playing cards (but not the jokers). However, that may not be desired, so these check-boxes can limit which card types have the watermark.

5.3.4 Card Item Priority

On rare occasions the Card Item order may need to be changed. For example, by default the Index is displayed on top of the Corner Pip when they overlap, however there may be a need to swap the order around. This list allows the order to be changed by selecting Card Items and using the buttons to move them in the list to create the order needed.

Note: the Card Item at the top of the list is displayed on top of all other Card Items, the item at the bottom is displayed underneath all others and those in the middle are similarly displayed relative to each other.

5.3.5 Jokers

When the 'Borderless Images' option is selected Joker images are displayed without a border to allow for a full card image.

5.4 Main Application Control Window - Colours Tab

Clicking on the 'Colours' tab shows the SVG colour settings.

5.4.1 Select Standard Court Colours

This panel selects which Standard image components colours should be modified by either the 'Colour Select' or 'Colour Set' controls. Using the mouse to click on the rectangular swatch selects a component to change. Holding down the control key and clicking allows multiple components to be selected and changed simultaneously.

5.4.2 Colour Select

The 'Colour Select' provides a colour ring to select the desired 'Hue' which can then be adjusted using the 'Saturation' and 'Brightness' sliders below. Alternatively the 'Red', 'Green' and 'Blue' sliders can be used. The 'Opacity' slider allows for components to be semi transparent.

5.4.3 Colour Set

If a specific colour value is required for an image component, the 'Colour Set' panel can be used. Either RGB or HSB values can be specified in either decimal or hexadecimal. Additionally, the scale for text box can be selected. For example, if a value is given as a percentage, '100' should be selected. RGB values are typically given in decimal in the range 0 to 255. Hue is typically given in decimal in degrees from 0 to 360.

Once a colour has been specified it must be set with the 'Apply' button.

5.5 Sample Navigation

The 'Sample Navigation' controls which card is displayed in the 'Sample' window, (except for Jokers).

5.6 Generate

The big blue button in the top right hand corner is the 'Generate' button. Clicking the 'Generate' button will create the cards in the 'Output' directory, based on the current selections.

5.7 Status Line

The Status Line updates the User when certain keys are pressed such as the Shift key or Alt key. The Status Line updates the User with information such as when settings are loaded or where card images are generated.

The Status Line also shows a progress bar when generating card images.

5.8 Sample Window

The 'Sample' window also provides some controls for resizing and positioning card items when it has the focus. A small green box indicates which card item has been selected for modification and is also used as a handle.

Note: the 'Modify Selected Card Item' control is updated to reflect any changes.

5.8.1 Mouse Controls

- Clicking in the ‘Sample’ window, but not on the handle, will cycle through the available card items.
- Dragging the handle with the mouse adjusts each instance of the selected card item.
- Scrolling up or down with the mouse wheel will increase or decrease the size of each instance of the selected card item.

Additionally the ‘Sample’ window can be moved by clicking in the window, but not on the handle, and dragging the window.

5.8.2 Keyboard Controls

With the focus on the ‘Sample’ window the following key presses can be used to modify the card items:

- The tab key will cycle through the available card items to indicate the currently selected item.
- The cursor keys move each instance of the selected card item.
- The ‘+’ and ‘-’ keys will adjust the size of each instance of the current card item appropriately.

Additionally the ‘Sample’ window can be moved by holding down the Shift key and using the cursor keys.

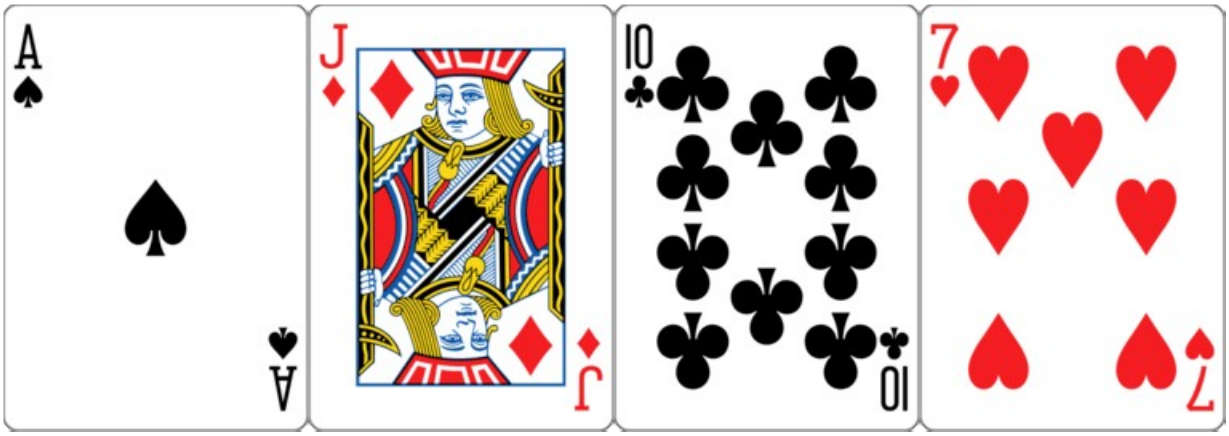
6 Generate Decks using the ‘CardWork’ environment

We will now explore the interface by creating examples.

6.1 Generate the Default deck

The following example uses the default parameters. After launching and clicking the Generate button, ‘CardCreate3’ uses default settings, taking SVG images for standard ‘faces’, ‘indices’ and ‘pips’ with card images generated in the output directory ‘cards/1’.

Sample output:

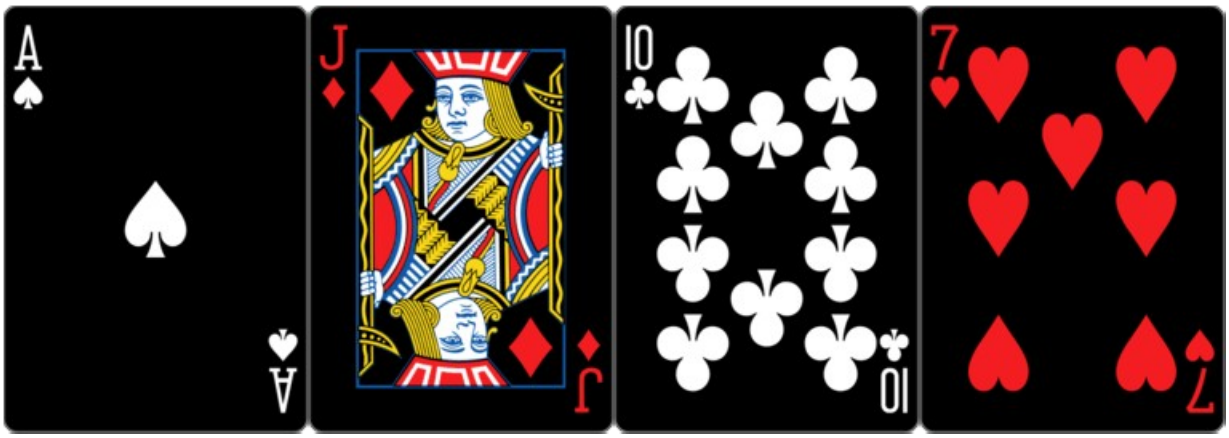


The following examples demonstrate how some of the controls are used.

6.2 Generate the Black deck

Here we select ‘Black’ as a ‘Colour Theme’ to change the colour of the card, pips and indices.

Sample output:



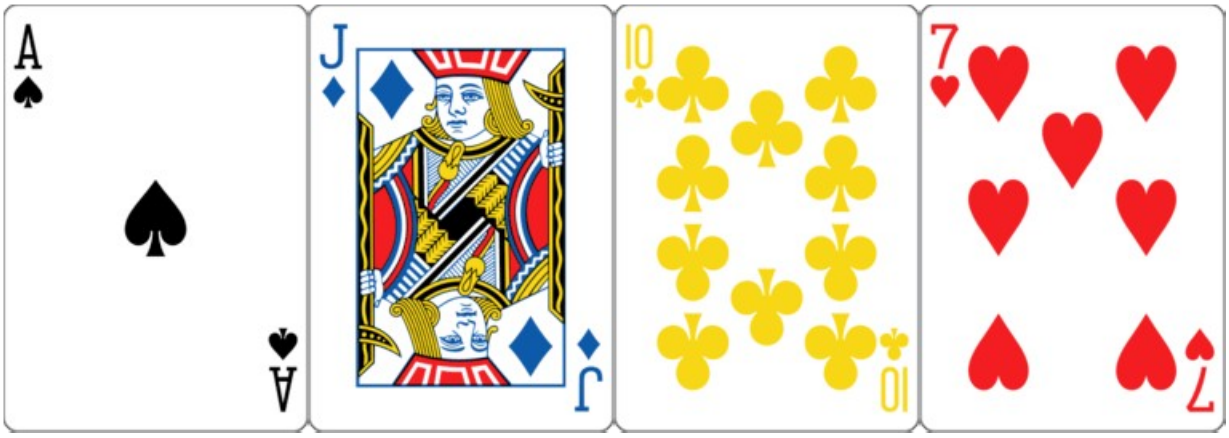
6.3 Generate the 4-colour deck

This example uses 4 colours for the pips and indices, but we utilize the blue and yellow of the face cards. However, to avoid the output being generated in the 'cards/1' directory, overwriting the contents we specify output to 'cards/4' by clicking the 'Manual' button and entering '4' in the now editable text box. This example illustrates the advantage of removing the pips from the standard court face images and adding them during generation so they match the pips used for the rest of the suit without the need to edit the court images.

For this example set the following non default values:

- From the Primary tab
    - Manually set the output directory (to '4' for example)
- From the Colours tab
    - Click on the 'Courts Yellow' swatch
    - Control click on 'Club Indices'
    - Control click on 'Club Pips'
    - Click on the 'Courts Blue' swatch
    - Control click on 'Diamond Indices'
    - Control click on 'Diamond Pips'

Sample output:



6.4 Generate the Rouen deck

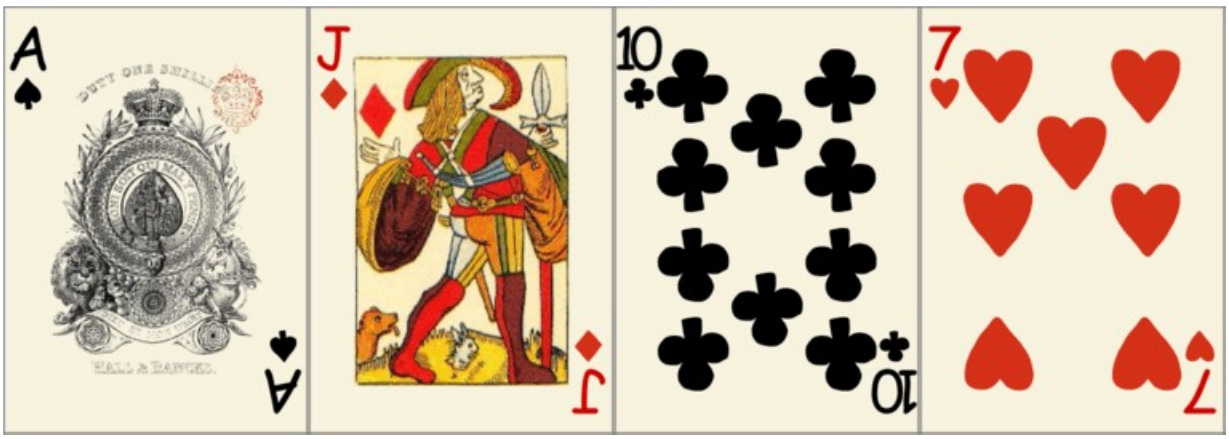
This example uses custom 'Face', 'Index' and 'Pip' styles to create a simulation of a Rouen deck. In this case the court card images are full length with pips, so we need to disable the generation of the pips on the court cards.

For this example set the following non default values:

- From the Primary tab
    - Deselect 'Use Standard Faces', 'Use Standard Indices' and 'Use Standard Pips'

    - ‘Face’, ‘Index’ and ‘Pip’ styles to ‘Rouen’
    - Deselect ‘Display Face Pip’
- From the Additional tab
    - ‘Corner Diameter’ to 0%
- From the Colour tab
    - ‘Background Colour’ to a custom RGB of 248, 243, 223

Sample output:



6.5 Generate Jumbo index

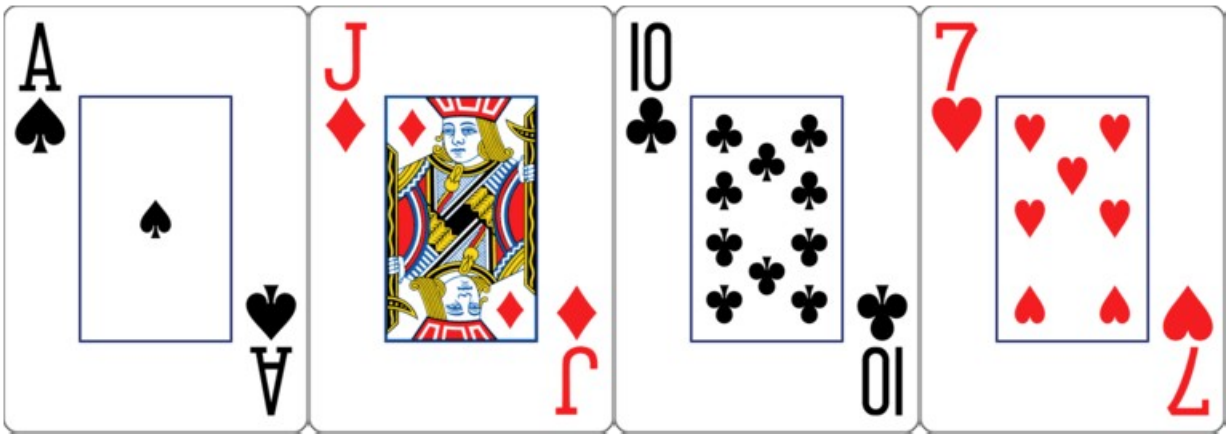
In this example we use standard ‘Faces’, ‘Indices’ and ‘Pips’, but modify several settings to adjust the size and position of the images to create an approximation of Jumbo index cards. We have also copied ‘JumboWatermark.png’ into the ‘faces/1’ directory and renamed it to ‘Watermark.png’. This will create a border in the background for all cards in this example. However the watermark is simply an image that is displayed in the background of all cards and could be any design.

Here is a breakdown of the values used in the ‘Modify Selected Card Item (%)’ for each of the card items:

	Indices	Corner Pip	Standard Pip	Face Pip	Face Image
Size	16	13	9	9	
X Centre	12	12	36	33.9	25
Y Centre	12	28	30	28.7	10

Additionally, to avoid overwriting the generated card deck in ‘cards/1’, manually set the output directory to ‘Jumbo’ (for example)

Sample output:



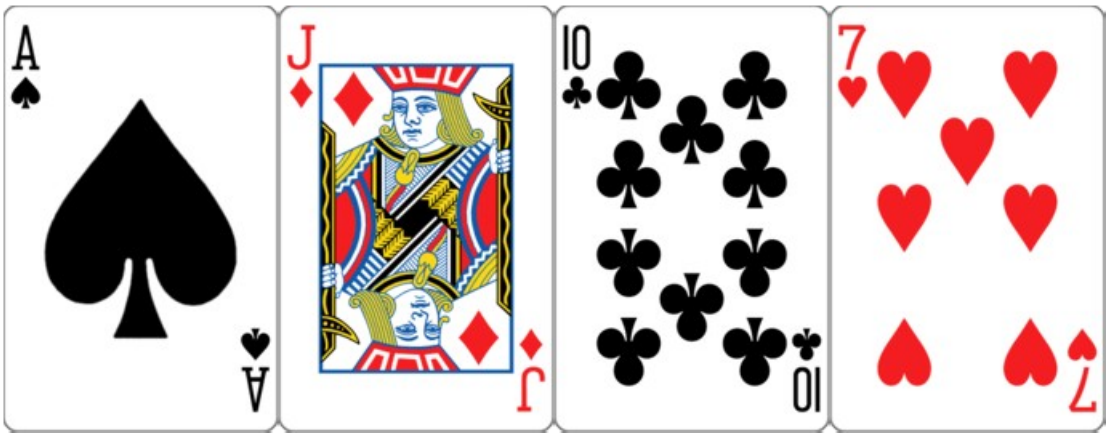
6.6 Generate the bridge size deck

So far, all cards have been created as poker decks. Poker decks are 3½ inches by 2½ inches. The default width is 380 pixels and the default height is 532 pixels. Both of these can be changed using the 'Card Size' parameters. The narrower Bridge cards are also 3½ inches high, but 2¼ inches wide, making it easier to hold more cards. Selecting the 'Bridge' radio button sets the correct width and height and maintains the aspect ratio if the height is adjusted.

Set the following non default values:

- Manually set the output directory to 'Bridge' (for example)
- 'Standard Pip' Size to 16% and the position to (27%, 18.7%)
- 'Face Pip' to a Size of 13% and a 'Y Centre' of 22%
- Select 'Use Image file if available'

Sample output:



7 Generate Decks using Add-ons

Now we generate decks that use an image on every card.

7.1 Generate the Terry Pratchett deck

This first example uses book covers from Terry Pratchett books. See the 'Packages' section (at the end of this document) for more details on packages. These packages are not currently available, however, they can be created manually.

After restarting 'CardCreate3' set:

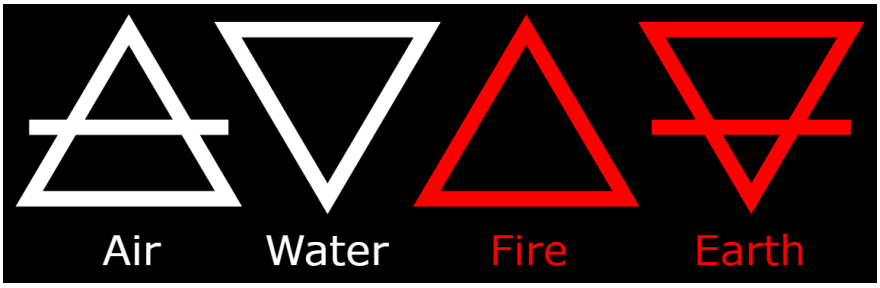
- Deselect 'Use Standard Faces'
- 'Face' style to 'Pratchett'
- Select 'Use Image File if available'
- Deselect 'Display Face Pip'
- Optionally select the Black 'Colour Theme'

Sample output:

7.2 Generate the Angels & Demons deck

Now an example that uses custom pips and custom images for the aces. This deck is based on Dan Brown's "Angels & Demons" and uses the John Langdon created ambigram images from it for the aces, the advantage being that you can read the card which ever way up you are holding it.

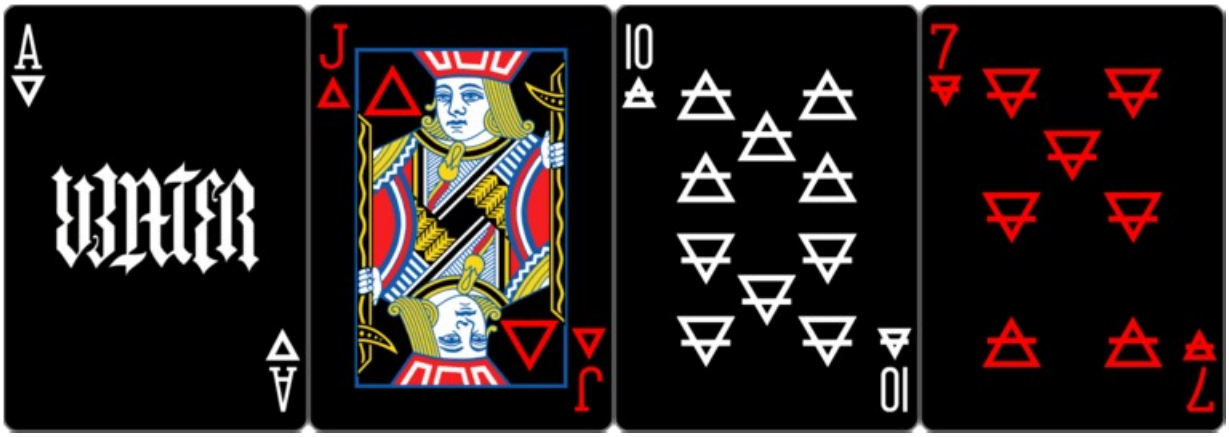
The deck also uses the ancient symbols for earth, air, fire and water. Angels are represented by the white pip cards of air and water. Demons are represented by the red pip cards of fire and earth.



After restarting ‘CardCreate3’ set:

- From the Primary tab
    - Deselect ‘Use Standard Faces’ and ‘Use Standard Pips’
    - ‘Face’ style and ‘Pip’ style to ‘AAD’
    - Select the Black ‘Colour Theme’
    - ‘Standard Pip’ Size to 12% and the position to (30%, 21%)
    - ‘Face Pip’ Size to 12% and the position to (27.5%, 20%)
- From the Colour tab
    - ‘Diamond Indices’ and ‘Heart Indices’ to RGB 255, 0, 0

Sample output:



8 Miscellaneous Examples

These decks illustrate other possibilities.

8.1 Custom Deck

To create this example deck with the style name of ‘Custom’ (for example), take the following actions:

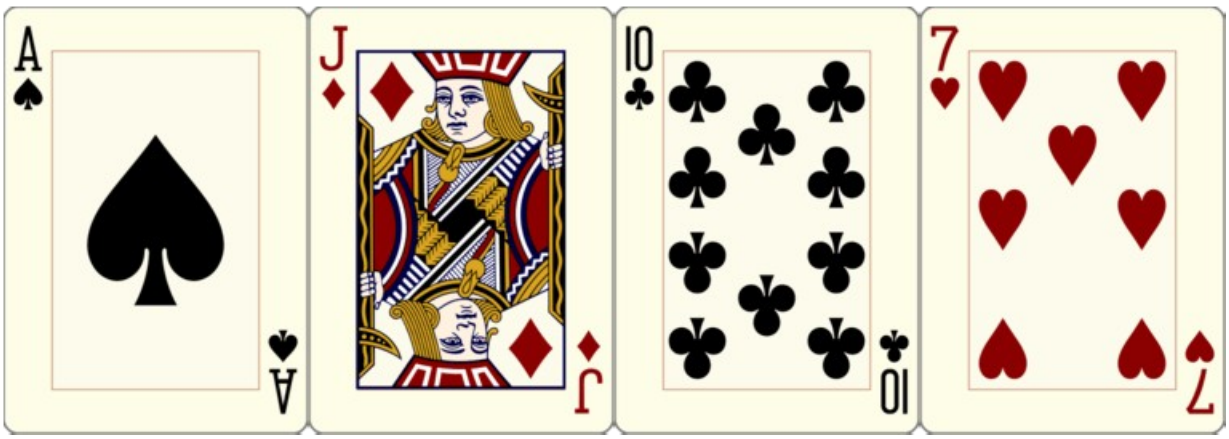
- Copy ‘Watermark.png’ from the boneyard into ‘faces/1’ directory
- ‘Output Directory’ to “Custom”
- Select the ‘Custom’ Colour theme
- Adjust the size of the standard pips to 14.3% and the position to (27.15%, 19.9%)
- Set the background colour to an RGB of 253, 253, 233

However, for the aces we will generate another set of cards with large pips:

- Adjust the size of the standard pips to 40%
- ‘Output Directory’ to “aces”

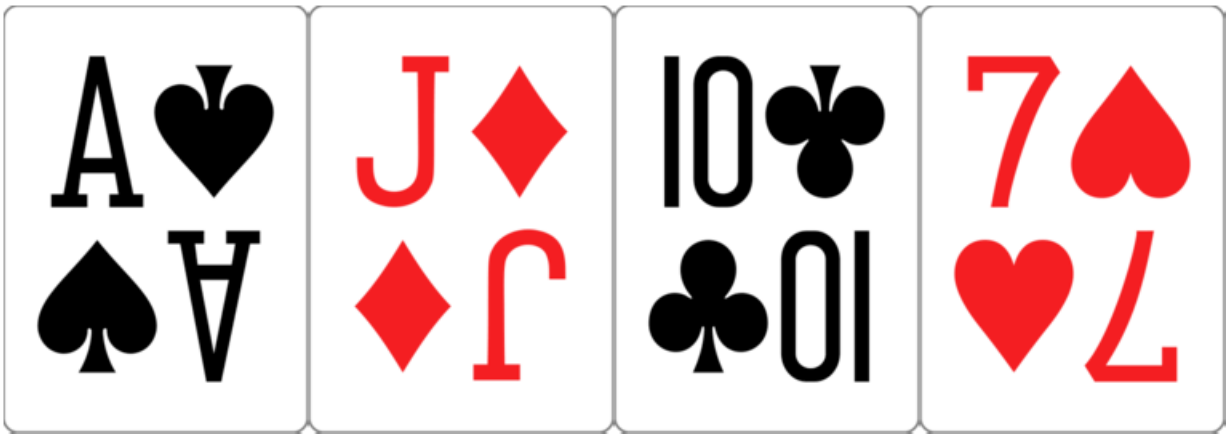
Copy the aces from ‘cards/aces’ onto ‘cards/Custom’.

Sample output:



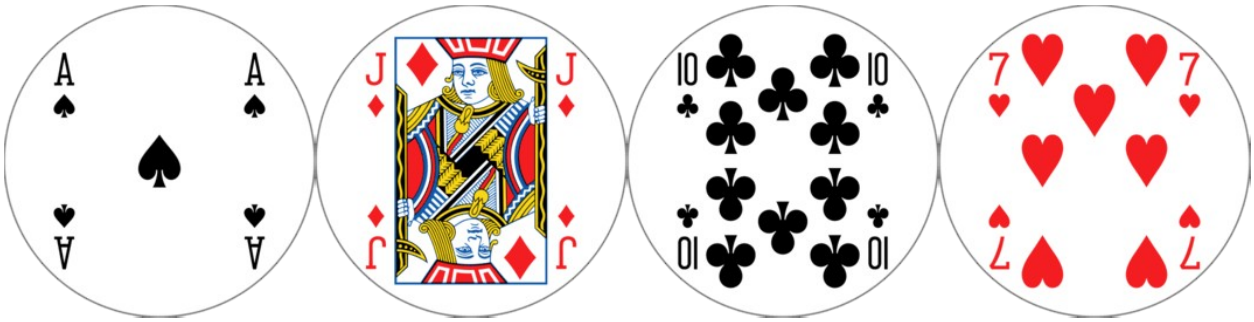
8.2 Minimalist Deck

For a plain minimalist deck, deselect displaying the standard pips, face images and face pips, increase the size of the indices to 40% and corner pips to 38%, in this example, and reposition them:



8.3 Round Deck

The cards round corners can be set excessively, in this example to 100%. Setting ‘Card Size’ to ‘Free’ and the card width and height to the same value and with some adjustments to the card items we get:



9 Notes

9.1 Image file type

All images **must be** .png files. It is a common and popular format that supports transparency. The use of transparency is highly recommended

9.2 File naming convention

The image files for the faces and indices use the same simple naming convention, which is the initial letter of the suit followed by the index or face image it represents. 'C' represents Clubs, 'D' for diamonds, 'H' for hearts and 'S' for Spades. For example, the face card image for the jack of diamonds is 'DJ.png' and is found in the 'faces' directory.

The naming convention for the pips image files is simply the suits initial letter only, so the pip for clubs is defined in the file 'C.png'. However, additional pip images can optionally be provided specifically for the court card faces or the corner pips. For additional court card face pips add of an 'F' after the suit letter, for example, a custom court card pip for diamonds is named 'DF.png'. Similarly for additional corner pips add an 'S' (for small) after the suit letter, for example, a custom corner pip for diamonds is named 'DS.png'.

9.3 Avoiding unnecessary file duplication

Typically the index for clubs and spades are black, so the index 'CA.png' will be used for the ace of clubs, but if 'SA.png' is not found 'CA.png' will also be used for the ace of spades. More generally, if images for indices for spades are not found, clubs will be used because they are typically the same colour. Similarly, if indices for hearts are not found, indices for diamonds will be used. This avoids unnecessary image duplication when suits use the same colour.

9.4 Packages

Packages are collections of custom images contained in a tar file. Typically they contain alternate face images, but may additionally include custom pips and/or indices also. New sub-directories are created in the 'faces', 'pips' and 'indices' directories, as needed, containing the new images. On start up, 'CardCreate3' includes these new sub-directories in the choice-boxes of the 'Input Directories' control pane. Packages are currently unavailable, so this must be done manually by creating the sub-directories and adding appropriately named images.